

Detector Readout Timing Study

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Darklight Software Meeting

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Question

Is the delay between the GEM and trigger readouts consistent between events and runs?

Motivation

If the uncertainty is too large, it may be difficult to find correlation between cluster and trigger timing in tracks.

We can look at the MPD and TDC clock values recorded in each event

MPD Clock

- 40 MHz $\rightarrow$ 25 ns increments
- There are four MPDs
- Related to when each GEM is read

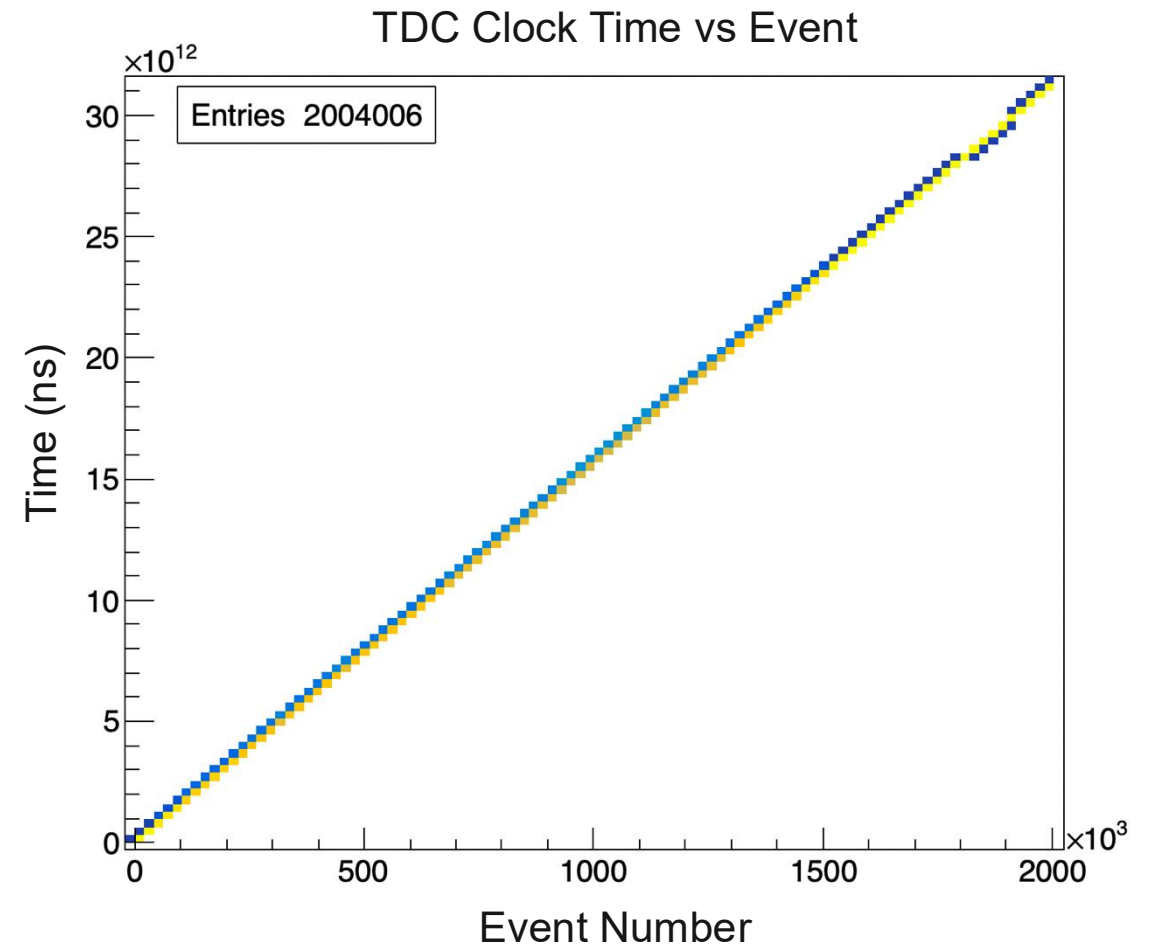
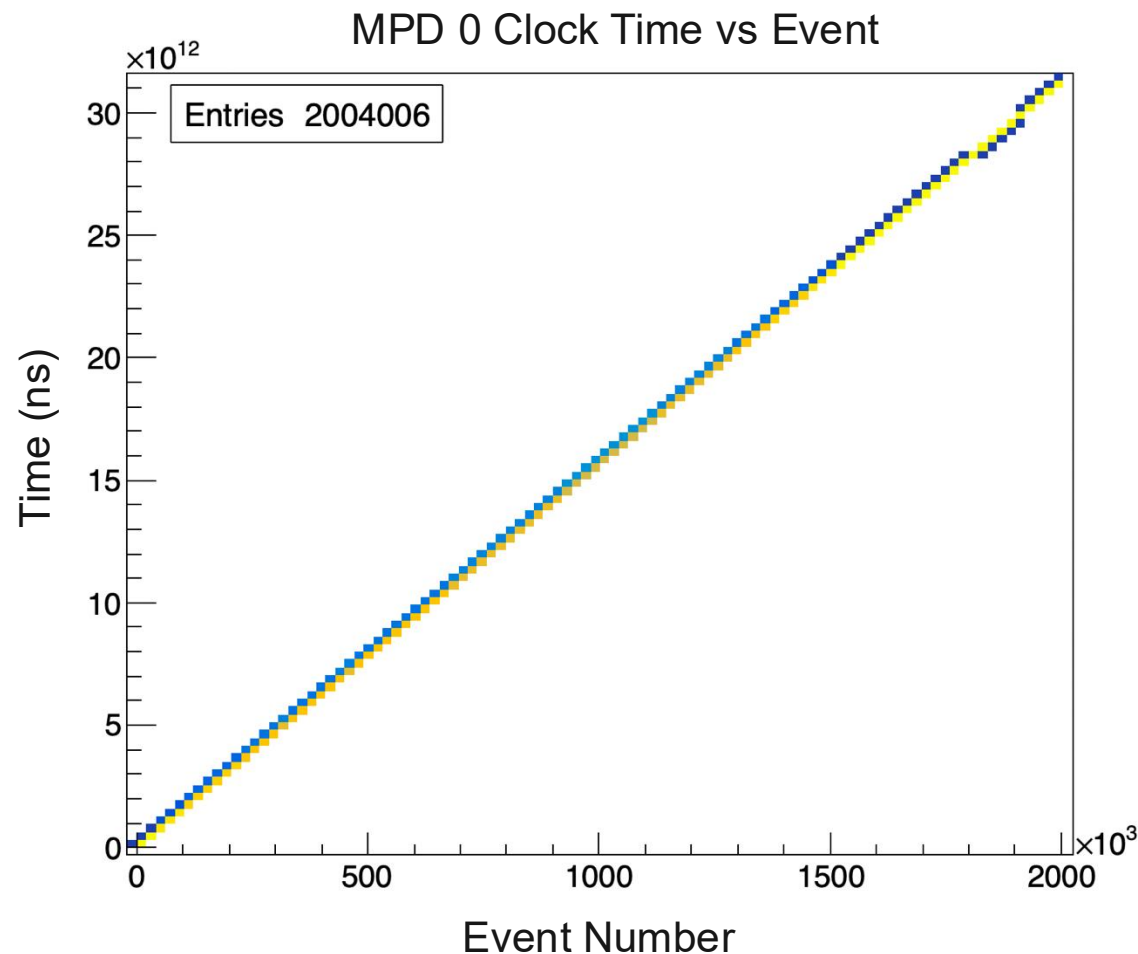
TDC Clock

- 100 MHz $\rightarrow$ 10 ns increments
- Related to when the triggers are read

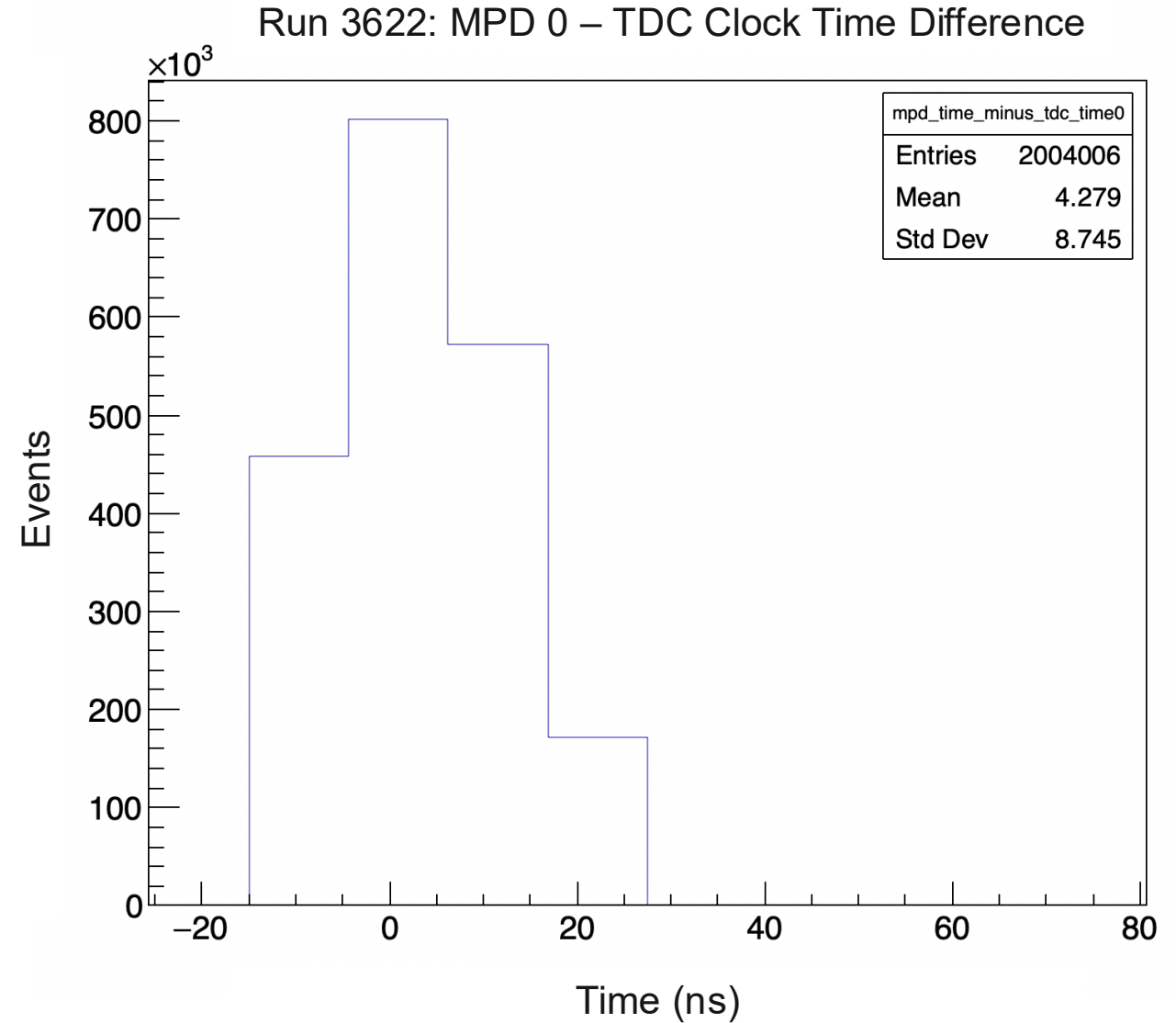
Converted raw clock times to relative timestamps

- Subtracted initial values
- Accounted for integer overflows

Run 3622
Cosmic Rays

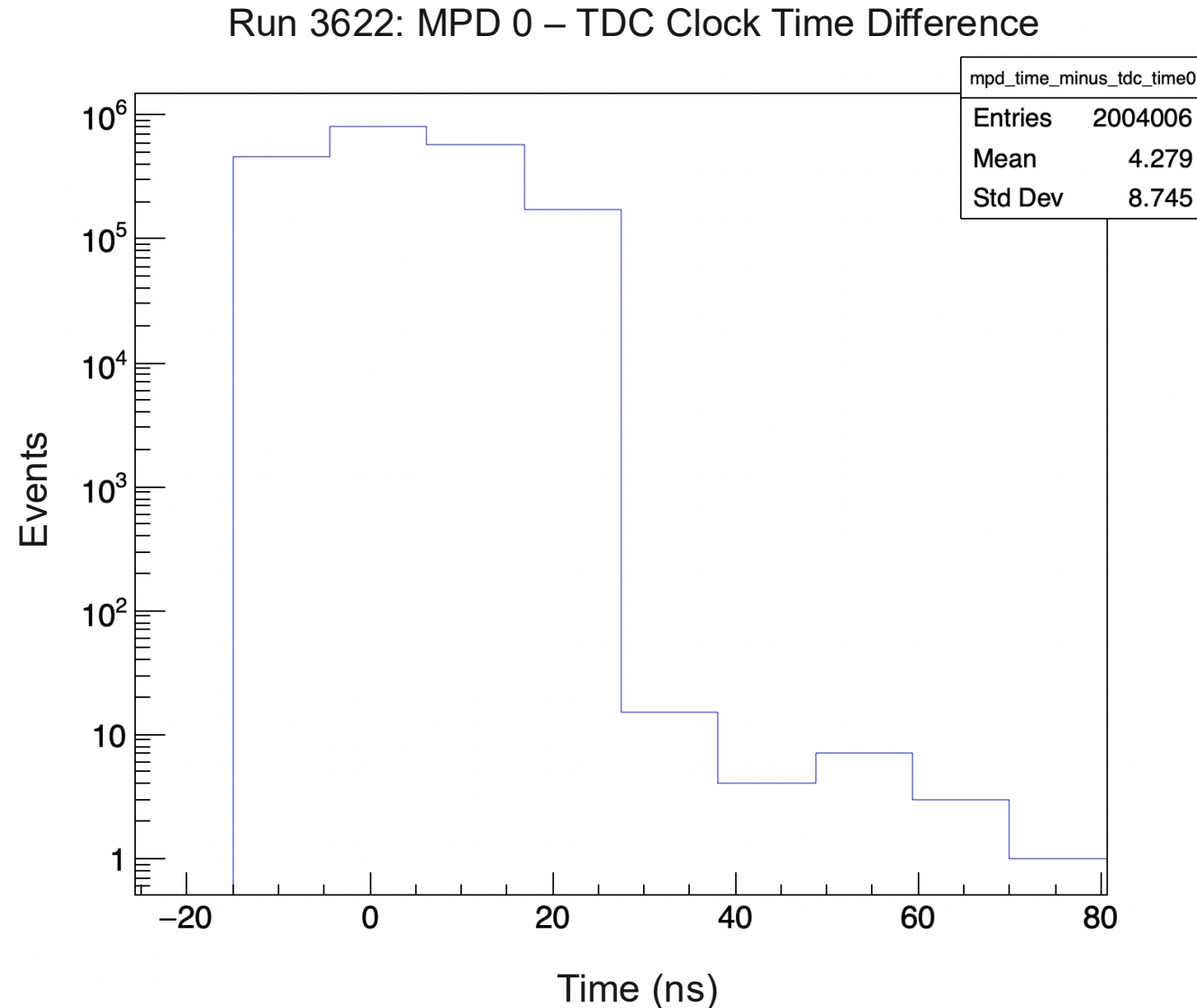


Difference in clock times is roughly constant across events in a run



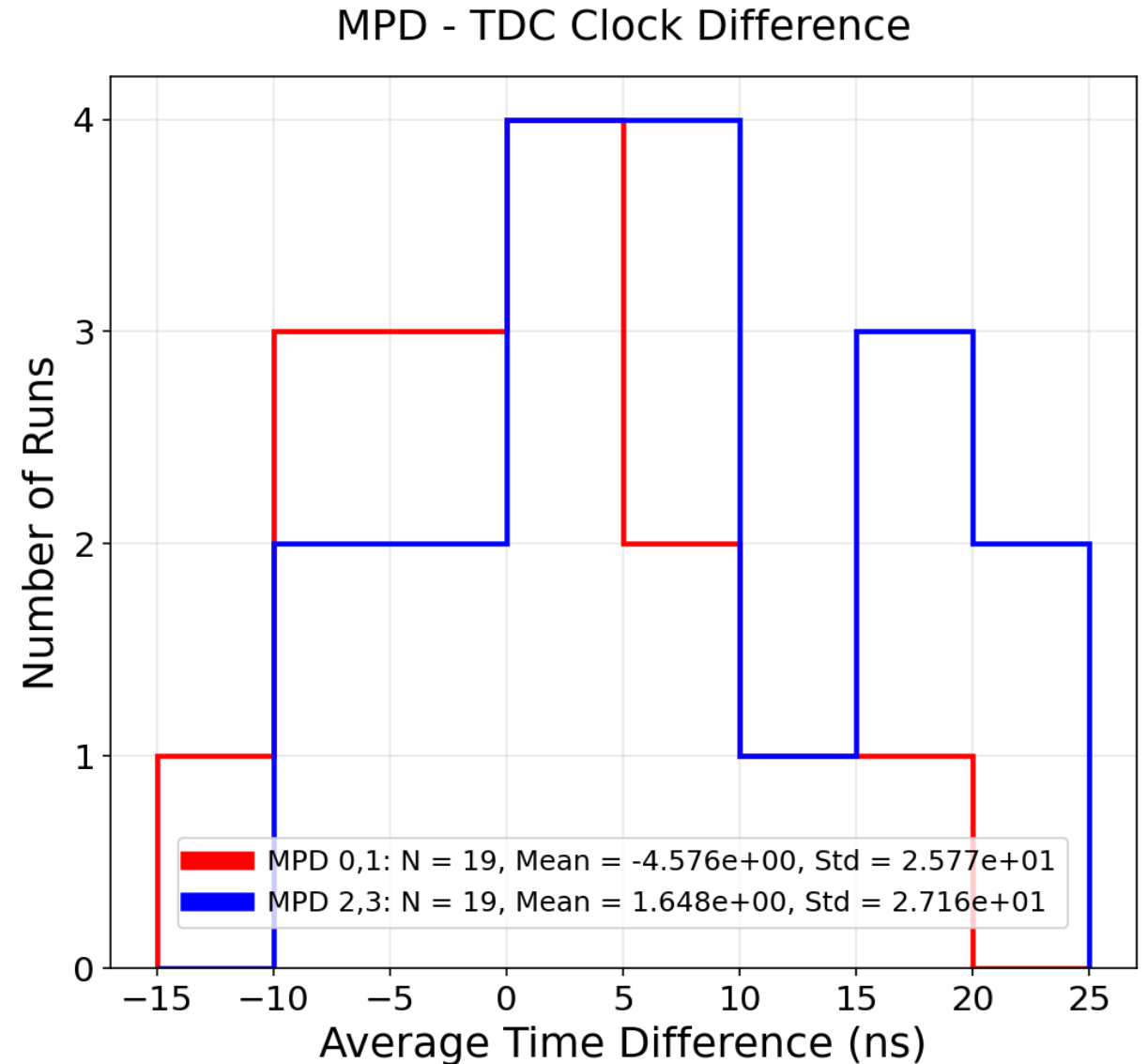
Difference in clock times is roughly constant across events in a run

- Expect ~ 35 ns range due to resolutions of clocks
- In this run we see 95 ns range
- Uncertainty in clock speeds?



Average difference in clock times is roughly constant across runs

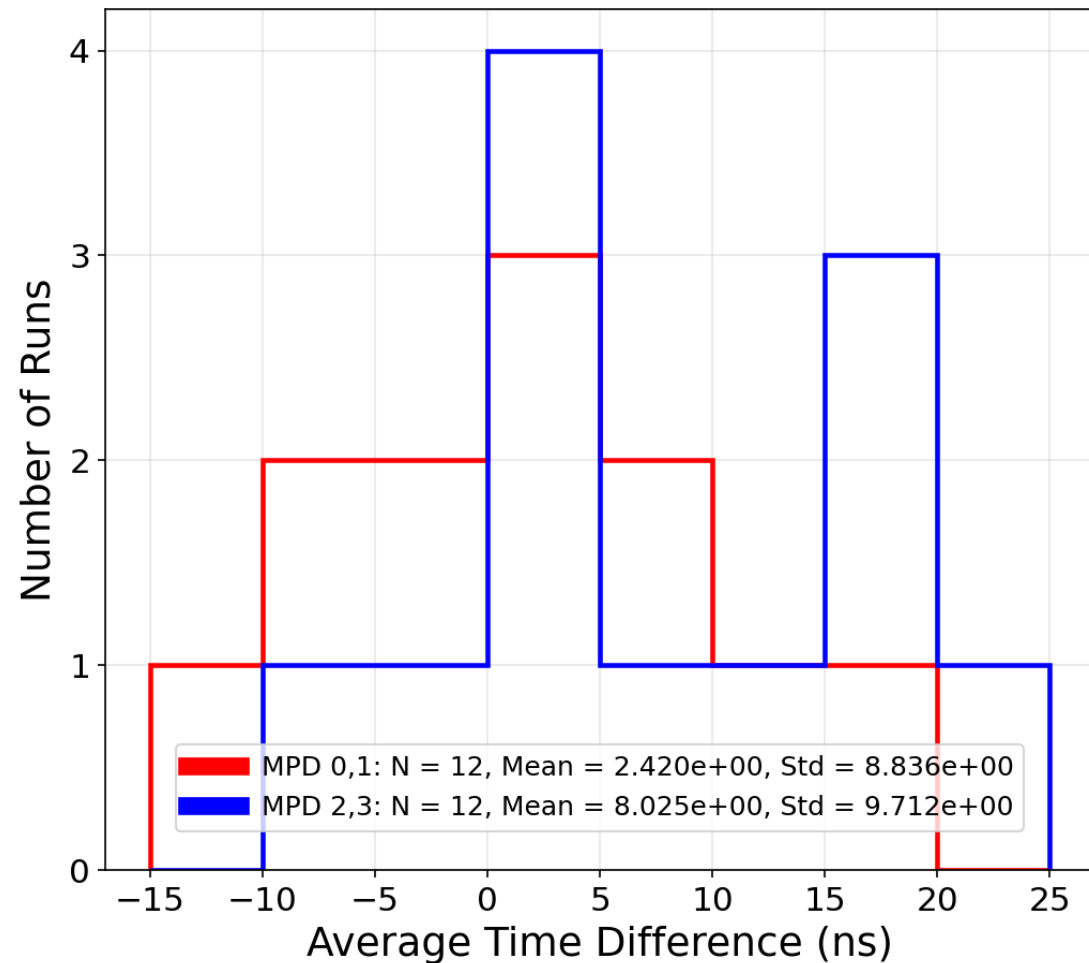
- 17 cosmic runs + 10 beam runs
- Range of values is mostly expected from clock resolutions
- Need more statistics to see any difference between cosmic and beam runs
- Left: -4.6 ± 5.9 ns
- Right: 1.6 ± 6.2 ns



Backup

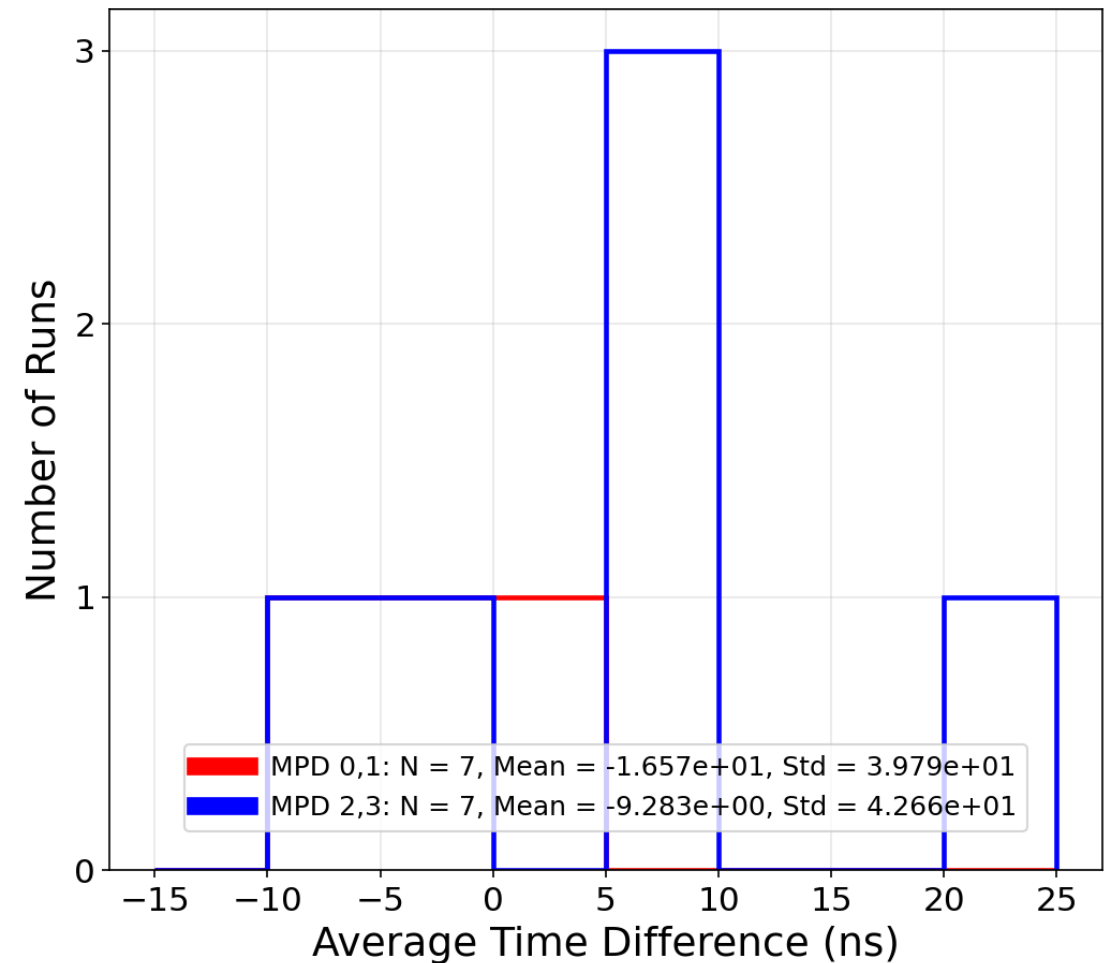
Cosmic runs

MPD - TDC Clock Difference



Beam runs

MPD - TDC Clock Difference



Immediate Goal: Correlate timing of clusters and triggers to improve track selection.

How do we do that?

- Determine “time” of clusters – look at ADC values over readout samples
- Look at clean, cosmic ray tracks
- Determine Δt between triggers and clusters in known tracks
- Select future tracks using this known time difference

Can we use Δt determined in cosmic data to reconstruct tracks in other runs?

How do we know that this Δt is the same across runs?

- Is the latency which we tuned constant?
- Are the GEM and trigger response times consistent?

Initial sanity check

- Verify that delay between GEM and trigger readouts is consistent

Average difference in clock times is roughly constant across runs

